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SECI1143 - PROBABILITY & STATISTICAL DATA ANALYSIS

SECTION 02

PROJECT 2 (INFERENTIAL STATISTICAL ANALYSIS)

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1.0 INTRODUCTION

1.1 Study Background & Purpose

Public transportation is a core part of daily life for commuters in Malaysia. This project uses daily ridership data from the government portal data.gov.my, which records the number of passengers using different types of public transport from January 1, 2019, to mid-2026. The main purpose of this study is to apply statistical analysis to this dataset. By doing so, we can move beyond simple observation and uncover the actual patterns and habits of people using the urban rail networks.

1.2 Objectives of the Study

This project aims to run statistical tests to better understand passenger behavior. The specific objectives are:

- Test if there is a real difference in the average daily ridership between the Ampang LRT Line and the Kajang MRT Line using an Independent T-Test.
- To see how closely the daily passenger numbers of the LRT Kelana Jaya Line and the MRT Kajang Line move together using Correlation Analysis.
- Create an equation that predicts the daily ridership of the LRT Kelana Jaya Line based on the Ampang LRT Line numbers using Regression Analysis.
- Checking if passenger traffic on the LRT Kelana Jaya Line is evenly spread out across all seven days of the week using a Goodness of Fit Test.

1.3 Research Scope & Expected Outcomes

This report focuses on the daily passenger numbers for three specific lines, the Ampang LRT, Kelana Jaya LRT, and Kajang MRT. The data covers a long period from 2019 to 2026, which gives us plenty of daily records to work with. At the end of this study, we expect to clearly show the differences in demand between these major train lines. We also expect to prove that passenger numbers are not just random, but follow a predictable weekly routine that links these transit systems together.

2.0 DATASET DESCRIPTION & PRE-PROCESSING

2.1 Overview of the Dataset

The empirical foundation of this data analysis utilizes public transportation data provided from data.gov.my website, which is the *Daily Public Transport Ridership* across Malaysia. This dataset records the daily commuter volume across various public transport types such as KTMB Shuttle Tebrau, Prasarana Bus such as *Rapid Penang*, *Rapid KL*, etc that are collected

in the span of January 1, 2019 through mid-2026.

2.2 Definition of Parameters

The primary parameters analysed are the **Date**, which serves as chronological index formatted as YYYY-MM-DD, and several quantitative columns which are the passenger volumes. For the hypothesis testing involving Two-Sample Independent T-Test, the variable *rail_lrt_ampang* and *rail_mrt_kajang* are utilised to conduct a proper hypothesis testing on the variance. For the correlation analysis, the daily ridership of the LRT Kelana Jaya Line and the daily ridership of the MRT Kajang Line are parameters tested for this analysis. Lastly, for the regression analysis, the Ampang / Sri Petaling LRT and the Kelana Jaya LRT parameters are used to conduct and analyse in this study.

3.0 DATA ANALYSIS

3.1 Hypothesis Testing (Two-Sample Independent T-Test)

3.1.1 Objective of the Test

The primary objective of this hypothesis testing is to conduct an independent two-sample t-test to investigate whether there is a statistically significant difference in the population mean (μ) daily ridership between the Ampang LRT Line and the Kajang MRT Line. By analyzing the raw data extracted from the *ridership_headline.csv*, this test aims to evaluate whether the observed variations in daily passenger volume reflect true structural differences in transit line demand in Malaysia, or merely just random daily fluctuations.

3.1.2 Variable Definitions

In this two-sample independent t-test, two primary variables are defined. The **Independent Variable (IV)** is the operational transit line classification, which acts as qualitative grouping for two distinct categories. The Ampang LRT Line & the Kajang MRT Line, measured in **nominal scale**

In other words, the **Dependent Variable (DV)** is the daily passenger volume that is measured by number of boardings per day, serving as **ratio scale**, which it contains a true mathematical zero

3.1.3 Mathematical Assumptions (Variance F-Test)

A two-sample F-test for equal variances is executed on the daily ridership records of Ampang LRT (σ^2_{Amfang}) and Kajang MRT (σ^2_{Kajang})

H₀ (null hypothesis) : $\sigma^2_{Amfang} = \sigma^2_{Kajang}$

H₁ (alternative hypothesis) : $\sigma^2_{Amfang} \neq \sigma^2_{Kajang}$

If the F-test calculation produces a p-value lower than the alpha threshold ($\alpha = 0.05$), the null hypothesis is rejected. We are going to execute a standard independent t-test to compute an accurate signal-to-noise ratio

3.1.4 Statistical Results & Output Interpretation

```
> var.test(lrt_ampang, mrt_kajang)

F test to compare two variances

data: lrt_ampang and mrt_kajang
F = 0.541, num df = 2676, denom df = 2676, p-value < 2.2e-16
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.5015135 0.5836001
sample estimates:
ratio of variances
 0.5410022
```

Figure 1: Variance test conducted in RStudio for Two-Sample test

The test produced an F-statistic of $F = 0.541$ with degree of freedom $df_1 = 2676$ and $df_2 = 2676$. The resulting significance values were $p = 2.2 \times 10^{-16}$. Because this p-value ($< 2.2 \times 10^{-16}$) is lower than 0.05, we **reject the null hypothesis (H_0)**. This shows that the daily ridership populations of the Ampang LRT Line and the Kajang MRT Line possess unequal variances. As a result, to preserve the integrity of Type I error rate, Welch's t-test is conducted in RStudio

```
> t.test(lrt_ampang, mrt_kajang, alternative = "two.sided", var.equal = FALSE)

Welch Two Sample t-test

data: lrt_ampang and mrt_kajang
t = -11.585, df = 4915.9, p-value < 2.2e-16
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -26588.34 -18891.84
sample estimates:
mean of x mean of y
 140353.3 163093.4
```

Figure 2: T-test conducted on RStudio for LRT Ampang & MRT Kajang

Based on the verification of unequal variances, a formal independent Welch two-sample t-test was conducted to evaluate population means. The sample data shown a mean daily ridership of Ampang LRT Line ($\bar{x}_{\text{Ampang}} = 140353.3$ passengers) and a higher mean for Kajang MRT Line ($\bar{x}_{\text{Kajang}} = 163093.4$ passengers). The inferential analysis produces a t-statistic of $t = -11.585$ with the adjusted degree of freedom, $df = 4915.9$. This results two-tailed significance value is $p < 2.2 \times 10^{-16}$, which falls below the alpha, $\alpha = 0.05$. The negative sign of t-value shows that the daily passenger load of the Ampang LRT Line is significantly lower than that of Kajang MRT Line. In addition, the 95% confidence interval between two populations spans from **-26.588.34 to -18891.84**. This clearly shows that the true mean daily ridership for

the Kajang MRT surpasses the Ampang LRT by an estimated margin of roughly 18892 to 26588 daily boarding

3.1.5 Statistical Conclusion

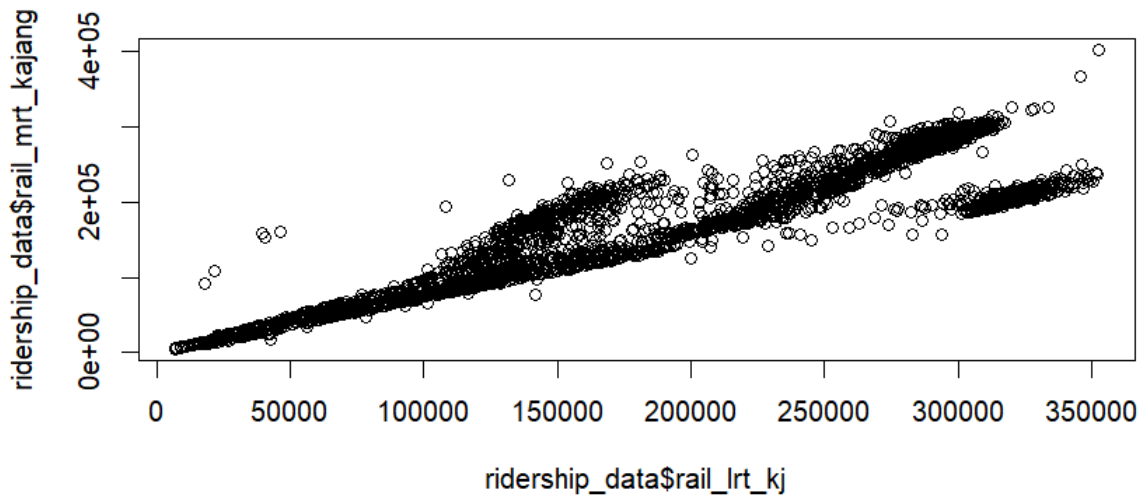
Based on the numerical output generated by the independent Welch two-sample t-test, a conclusion has been made. Due to the calculated p-value ($p < 2.2 \times 10^{-16}$) is smaller than the $\alpha = 0.05$, the null hypothesis is **H_0 is rejected**. When integrated with the calculated sample parameters, the mean daily passenger count of the Kajang MRT ($\approx 163,093$ boardings) exceeds Ampang LRT ($\approx 140,353$ boardings). It mathematically verifies that the Kajang MRT system handles a fundamentally heavier commuter load on average day. This variance indicates a distinct, structural difference in passenger volume & transit demand across two critical corridors within the public transport network

3.2 Correlation Analysis

3.2.1 Variables and objective

The variables evaluated in this analysis are the daily ridership of the LRT Kelana Jaya Line (the independent/predictor variable, plotted on the X-axis) and the daily ridership of the MRT Kajang Line (the dependent/outcome variable, plotted on the Y-axis), both measured as the total number of individual passenger boardings per day. The primary objective of conducting this statistical test is to determine the strength and direction of the linear relationship between the city's two most critical urban rail transit networks across a long-term timeline spanning 2019 to 2026. By assessing how closely these daily passenger volumes track together, the test aims to uncover whether commuter traffic patterns on one network reliably mirror, complement, or operate independently from the other, providing essential empirical data regarding regional transit dependencies.

3.2.2 Result and interpretation



The analysis yielded a Pearson correlation coefficient (r) of approximately 0.91, which indicates an extremely strong positive relationship between the daily ridership of the LRT Kelana Jaya line and the MRT Kajang line. Because the score is positive and very close to the maximum possible value of 1.0, it mathematically proves that the passenger volumes on these two primary rapid transit lines move in near-perfect synchronization. When commuter numbers spike on the LRT Kelana Jaya line (such as during standard working weekdays), they spike almost identically on the MRT Kajang line; conversely, during weekends, public holidays, or festive seasons, both lines experience the exact same sharp drops in passenger traffic.

3.2.3 Conclusion

In conclusion, this correlation analysis proves that the daily ridership demands of the LRT Kelana Jaya and MRT Kajang lines are deeply intertwined and share a highly predictable macro-behavioral pattern. While a strong correlation coefficient does not inherently imply a direct causal relationship—meaning a passenger spike on the LRT does not directly *cause* a passenger spike on the MRT—it conclusively establishes that both independent networks are responding to the exact same external socio-economic forces. The data clearly shows that a massive proportion of the urban population follows a unified, synchronized commuting routine. For transportation operators and urban planners, this near-perfect alignment means that any major event, service disruption, or capacity change affecting one line will almost certainly carry immediate, parallel implications for the passenger load and operational pressures of the other.

3.3 Regression Analysis

For this part we wanted to find out whether the daily ridership on one Klang Valley rail line can actually be used to predict the ridership on another, and simple linear regression (the whole of Chapter 7 Part 2) was the obvious tool for the task. If everyone in the Klang Valley is heading to work or campus on a Tuesday, the lines all fill up together, and if it is a public holiday they empty out together. So we expected the two lines to rise and fall more or less as a pair, and regression lets us put an actual equation to that hunch.

3.3.1 Choosing the variables

The dataset gives daily ridership for a whole list of rail and bus services, but a lot of those columns have missing values for the earlier years, especially the bus ones. We picked the two services that were recorded every single day across the entire 2019 to 2026 range: the Ampang / Sri Petaling LRT and the Kelana Jaya LRT. That gave us $n = 2708$ days with zero missing data, which is about as clean as real-world data is ever going to be.

We set the Kelana Jaya line as the dependent variable (y) and the Ampang line as the independent variable (x). To be honest the choice of which one is x and which is y is a little bit arbitrary here, because neither line really “causes” the other, they are both just driven by the same commuters. But the regression still holds up perfectly fine as a predictive relationship, and that prediction angle was what we were after.

3.3.2 Scatter plot and correlation

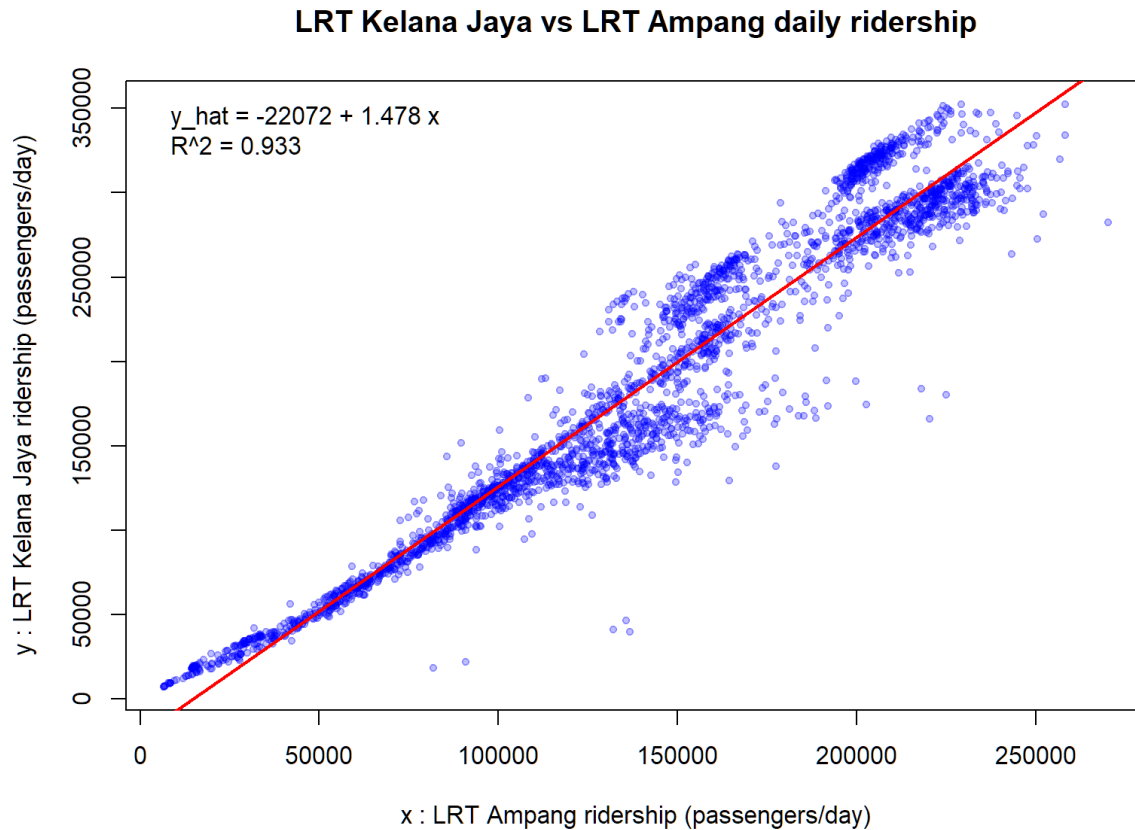


Image: Regression Scatter With Line

The first thing we did was just plot the two variables against each other. The scatter plot shows a clear upward band of points, where higher Ampang ridership lines up with higher Kelana Jaya ridership, and the cloud is tight enough that you can basically eyeball the straight-line relationship before fitting anything. To actually put a number on the strength of it we ran `cor.test()` in R and got a Pearson correlation of $r \approx 0.966$. That is very close to 1, so it is a strong positive linear relationship, and it came out significant with $p < 2.2e-16$.

3.3.3 The regression equation

summary(model) output:

```
Call:
lm(formula = rail_lrt_kj ~ rail_lrt_ampang, data = dat)

Residuals:
    Min       1Q   Median       3Q      Max
-140366  -11510    -914   11855   60727

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -2.207e+04  1.164e+03  -18.95  <2e-16 ***
rail_lrt_ampang  1.478e+00  7.594e-03  194.67  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 23820 on 2706 degrees of freedom
Multiple R-squared:  0.9334,    Adjusted R-squared:  0.9333
F-statistic: 3.79e+04 on 1 and 2706 DF,  p-value: < 2.2e-16
```

Next we fit the least-squares line with `lm()`. The estimated regression equation came out as:

$$\hat{y} = -22071.68 + 1.4784x$$

The slope ($b_1 = 1.478$) is the part we care the most about. For every extra passenger on the Ampang line the model predicts roughly 1.48 more passengers on the Kelana Jaya line. This implies that Kelana Jaya is a busier line overall.

The intercept ($b_0 = -22072$) is predict Kelana Jaya ridership for a day when Ampang line carries zero passengers, which is ignored due to out of scope of our data.

3.3.4 Goodness of Fit Test

After that we looked at how good the fit actually is. The coefficient of determination came back as $R^2 \approx 0.933$, meaning about 93% of the day-to-day variation in Kelana Jaya ridership is explained by the variation in the Ampang line. It also lines up with the correlation, because R^2 is just r^2 ($0.966^2 \approx 0.933$), exactly like the note on the slides points out.

The standard error of the estimate is 23,817 passengers. That is the typical gap between what the line predicts and the actual ridership on a given day.

3.3.5 Significance test on the slope (t-test)

```
> # ---- A5. t-test on the slope (H0: beta1 = 0 vs H1: beta1 != 0) -----
> coef(summary(model))      # Estimate / Std.Error / t value / Pr(>|t|)
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -22071.682007 1.164470e+03 -18.95427 2.503095e-75
rail_lrt_ampang 1.478402 7.594303e-03 194.67252 0.000000e+00
> confint(model)           # 95% CI for b0 and b1 (slope CI ~ [1.464, 1.493])
              2.5 %      97.5 %
(Intercept) -24355.023344 -19788.340669
rail_lrt_ampang 1.463511 1.493293
```

To make sure the relationship is not a coincidence, we ran the t-test on the slope from Part 3 of the chapter. The hypothesis are:

$$H_0: \beta_1 = 0$$

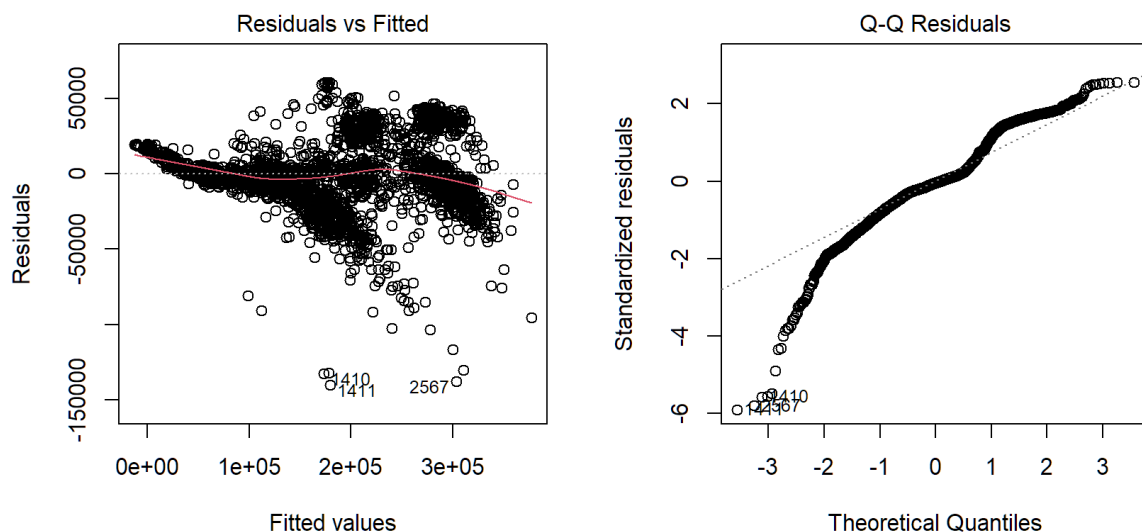
$$H_1: \beta_1 \neq 0$$

The test statistic, $t = 194.7$ and p-value below $2.2E-16$, so we reject null hypothesis at the 5% confidence level. Additionally the 95% confidence interval for the slope, $[1.46, 1.49]$ which does not contain 0, so it supports our conclusion. In conclusion there is a statistically significant linear relationship between ridership on the two lines.

3.3.6 Using the model to predict

For this part, we can utilize the equation to form an educated prediction given certain values. For example, if the Ampang line carried 150,000 passengers on some day, we can predict that Kelana Jaya ridership is about 199,700. However, stretching the line out to something like 0 or 500,000 would not be reliable, because the model was only ever fit on the range the real data covers.

3.3.7 Checking the assumptions



In order for a model to be considered valid, we checked the assumptions. We looked at the linear underlying relationship, independent errors, and errors that are normally distributed with constant variance. We did this with two residual plots that map straight onto those assumptions. The Residuals vs Fitted plot is for eyeballing linearity and constant variance, and ours sits as a fairly even band around zero, although it does fan out a little at the higher fitted values, so the constant-variance assumption is not flawless. The Normal Q-Q plot checks whether the errors are roughly normal, and most of our points sit nicely on the reference line with a bit of drift in the tails. Given how strong the relationship is, we are comfortable saying the regression is valid and genuinely usable.

3.4 Goodness of Fit Test

For the optional component we chose the goodness-of-fit test. We are testing whether ridership is uniform across days of the week. It's generally assumed that weekends feel quieter, but instead of just assuming it we figured we should actually test whether the weekday pattern is statistically real.

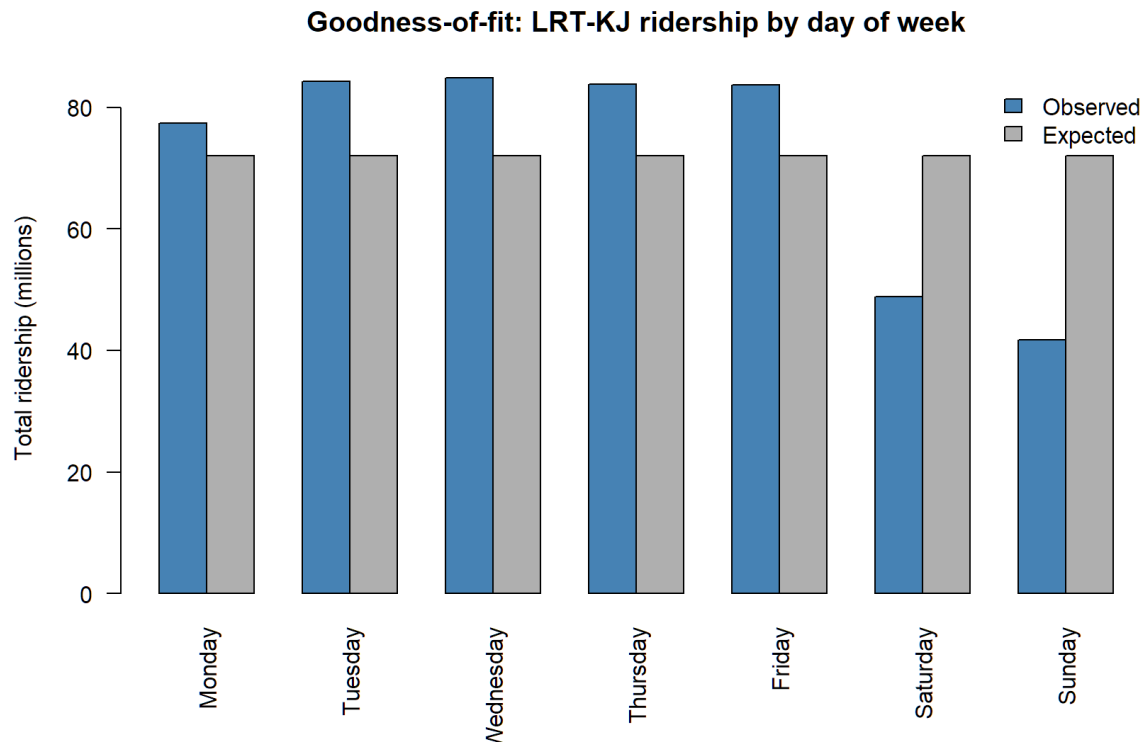
3.4.1 Test and the hypotheses

The goodness-of-fit test then compares those observed totals against the totals we would expect if ridership really were perfectly uniform across the week. The hypotheses are:

H_0 : ridership is uniformly distributed across seven days

H_1 : ridership is not uniformly distributed across days of the week

3.4.2 Observed vs expected



Under H_0 the expected total for each day is the total divided by 7, which equals to 72.1 million per day. However, the observed total does not show this pattern. . The five weekdays all sat somewhere around 77 to 85 million each, while Saturday dropped to about 48.8 million and Sunday fell all the way down to about 41.8 million.

3.4.3 Discussion

Our analysis results indicate that the LRT is noticeably emptier on some days than others. Especially two weekend days' expected ridership is significantly below uniform expectations.

4.0 CONCLUSION & REFLECTION

4.1 Summary of Key Findings

Based on the statistical tests conducted on the Klang Valley public transport dataset, we can draw several clear conclusions:

- Independent T-Test: The analysis proved that the Kajang MRT line handles significantly more passengers on an average day compared to the Ampang LRT line.
- Correlation Analysis: There is a very strong positive connection between the LRT Kelana Jaya and MRT Kajang lines. Their daily passenger numbers rise and fall in near-perfect synchronization.
- Regression Analysis: We successfully built a reliable predictive model showing that the Kelana Jaya line's ridership can be estimated based on the Ampang line's numbers, with the model explaining approximately 93% of the day-to-day variation.
- Goodness of Fit Test: The test confirmed that passenger volume is not spread evenly across the week. Expected ridership drops significantly during the two weekend days compared to the much higher traffic seen on weekdays.

4.2 Insights & Personal Learnings from Project Activities

Working on this project successfully bridged the gap between theoretical statistics and real-world application. By handling and processing actual public transport data, we saw how numbers can reveal clear, structured patterns in human behavior and city routines. The process highlighted the importance of careful data selection and pre-processing before running any tests. Furthermore, utilizing R-Studio to execute these analyses helped solidify our technical foundation, which is an essential step forward in our computer science and data engineering studies. Ultimately, this project demonstrated that statistical data analysis is more than just calculation but a vital tool for making informed, data-driven observations about the world around us.

5.0 APPENDIX

5.1 Sample of Original/Raw Dataset

Our team sourced the official data from the [Daily Public Transport Ridership portal on data.gov.my](https://data.gov.my) to ensure accuracy in our analysis. The latest .csv dataset can be downloaded and retrieved from https://storage.data.gov.my/transportation/ridership_headline.csv

5.2 Presentation Video Link

<https://youtu.be/A43AtltoztU>